Can We Do Interpretable NLI with Graphs Based on Atomic Propositions?

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Introduction

Does the Premise P logically entail, contradict or leave neutral the Hypothesis H?

P A man is playing a guitar on stage.
H A man is playing an instrument. — entailment
H The man is a professional musician. — neutral
H The man is asleep. — contradiction

Current NLI systems lack true reasoning and rely on surface-level pattern matching and annotation artifacts, making them uninterpretable black boxes. A hypothesis-only classifier reaches 67% on SNLI [1].

Our approach:

Auditable evidence: P and H become atomic knowledge graphs [2]; the classifier never sees the raw text.

Knowledge bridging: ConceptNet [3] edges (direct or two-hop) link the entities of P and H.

Example

Example — entailment

P The dog and the cat are in the kitchen.
H An animal is in a room.

1. Atomic propositions

P The dog is in the kitchen. The cat is in the kitchen.
H An animal is in a room.

2. Triples

P (dog, AtLocation, kitchen) (cat, AtLocation, kitchen)
H (animal, AtLocation, room)

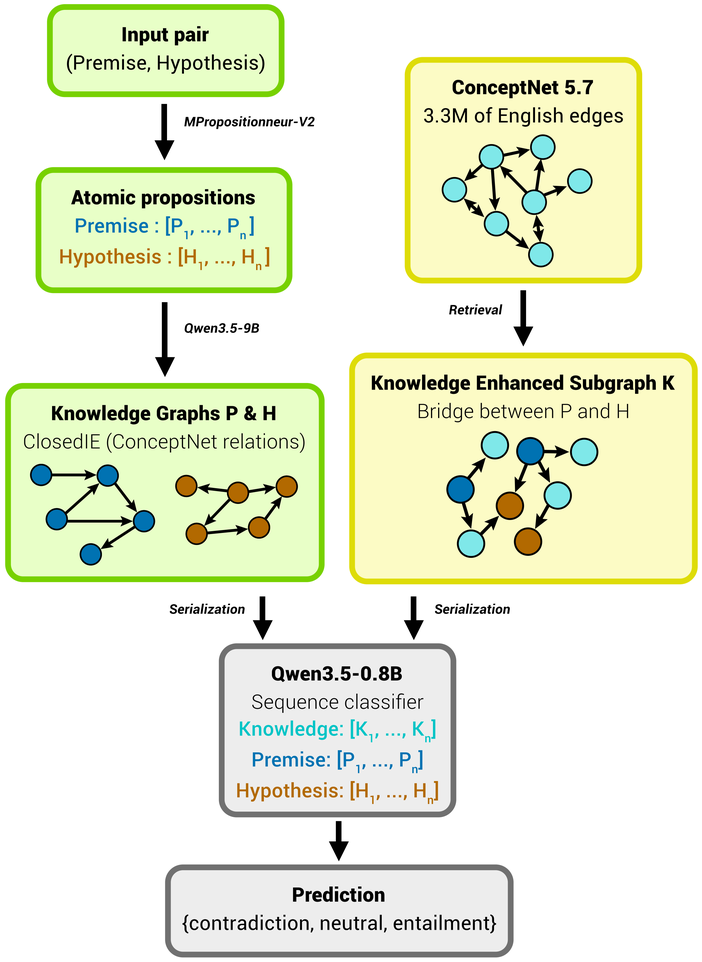
3. Retrieved knowledge

dog $\xrightarrow{\text{IsA}}$ animal kitchen $\xrightarrow{\text{IsA}}$ room

4. Closure

(dog, AtLocation, kitchen) $\wedge$ (dog, IsA, animal) $\wedge$ (kitchen, IsA, room) $\Rightarrow$ (animal, AtLocation, room)

The 2 edges of K make the missing link explicit.



Results

	Graph	Text	Price of interpretability
SNLI [4]	0.897 ± 0.006	0.916	−1.9
ANLI R1 [5]	0.565	0.704	−13.9
ANLI R2	0.462	0.572	−11.0
ANLI R3	0.463	0.553	−9.0

Accuracy of Qwen3.5-0.8B on serialized graphs vs. raw text. SNLI [4]: crowdsourced captions; ANLI [5]: adversarially collected in three rounds.

Limits

1. Triples lose modality

*The dog **or** the cat **might** be in the kitchen.*
$\rightarrow$ (dog, AtLocation, kitchen) (cat, AtLocation, kitchen)

or and *might* disappear: both animals are now in the kitchen.

2. Knowledge reflects statistical regularities

SNLI validation #4360 — contradiction

P A white race dog wearing the number eight runs on the track.
H A white race dog runs around his yard.

Retrieved knowledge

dog $\xrightarrow{\text{AtLocation}}$ backyard $\xrightarrow{\text{IsA}}$ yard

Dogs are usually found in yards, so ConceptNet links them. A noisy K can contradict P, support H and pull the model away from contradiction. On SNLI, removing K does not hurt (0.895 vs. 0.892).

Conclusion & Future Work

Graphs alone nearly match text on SNLI (−1.9 pts), but not on ANLI (−9 to −14 pts). The gap is representational rather than data-limited.

Next:

Fewer black boxes: parser-based extraction and a small graph classifier instead of LLMs.

Logical closure: enhance P using entailment.

References

[1] Gururangan et al. *Annotation artifacts in natural language inference data.* NAACL-HLT 2018.
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[3] Speer et al. *ConceptNet 5.5: an open multilingual graph of general knowledge.* AAAI 2017.
[4] Bowman et al. *A large annotated corpus for learning natural language inference.* EMNLP 2015.
[5] Nie et al. *Adversarial NLI: a new benchmark for natural language understanding.* ACL 2020.